Cross-Round Basin Dynamics of a Geometry-Only SHA-256 Reverse Tension Probe

The Two-Read-Head Cancellation: Why Depth Tightens the Basin Exactly as Much as It Widens the Search

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We study the reverse exploration of a single SHA-256 compression round under a *geometry-only* scoring function—a tension probe that scores a candidate message word by how well the carry geometry of the staged additions it induces matches the carry geometry of the true forward computation, without ever comparing the word to a stored target. We reconstruct, from scratch and on an independent bench, the central *Zero Score Theorem*: the true word, and only the true word among 2^{32} candidates, achieves a geometry score of exactly zero, while 200,000 random false candidates yield zero zeros. We then ask the question the probe's own authors flagged as open: *does the cold basin collapse when the constraint is chained across rounds?* Our central empirical result is a clean cancellation. When cross-round consistency is checked using the *true* predecessor word (an answer-key verifier), the cold basin collapses by a factor of $\approx 5\times$ per added round, projecting to unique recovery in ≈ 4 rounds. But when the answer-key is removed—when each added round must *find* its own predecessor word by the same cold criterion—the collapse vanishes: $300/300$ round-63 cold seeds admit a cold round-62 predecessor, a density collapse factor of exactly $1.00\times$. The depth axis that tightens the basin and the search axis that widens it are equal and opposite. We frame this as a *two-read-head* structure and argue it is a measured, local instance of complexity conservation: the constraint that collapses the frontier is itself as expensive to obtain as the collapse is valuable. We are explicit that this is *not* a SHA-256 inversion, and that the $1.00\times$ figure is the honest headline number, not the $5\times$ one.

Introduction and Scope

This paper occupies a deliberately narrow piece of ground, and the value of the result depends on keeping it narrow. We are *not* claiming an inversion of SHA-256. We are not claiming a preimage attack, a collision technique, or any reduction in the cryptographic security of the hash. What we are doing is measuring, on a live bench, the *shape* of a particular reverse-exploration landscape, and reporting a clean quantitative law about how that shape behaves when read in two directions at once.

The motivation is a question that arises naturally from the structure of the SHA-256 round function and was named as the central open problem in an earlier geometry-only tension probe . The forward round function is cheap: given the state and the message word, one computes the next state in a fixed number of operations. The reverse step, in isolation, is also cheap in a surprising sense—given the *next* state and a *candidate* message word, one can algebraically recover a unique predecessor state (Theorem 1 below). The difficulty of inversion does not live in any single reverse step. It lives in the fact that the message word is unknown, and that the criterion for recognizing the correct word, applied to one round in isolation, admits an enormous *cold basin* of false positives. The open question is whether *chaining* the criterion across rounds collapses that basin fast enough to isolate the true word.

We answer this question with a measurement, and the measurement has two faces. Read one way—with access to the true predecessor words—the basin collapses quickly. Read the other way—honestly, with each round forced to find its own predecessor—the basin does not collapse at all. The two readings are not in conflict; they are two read-heads dropped at two points on a single structure, and the gap between them is precisely the cost of the information that the first reading assumes and the second reading must pay for.

A note on method and authorship

The substantive framework, the tension-probe construction, and the guiding intuition that "depth collapses the basin" are due to the first author, whose multi-decade research program treats executable code as the sole arbiter of a structural claim. The bench reconstruction reported here—the independent re-derivation of the Zero Score Theorem, the two-read-head experiment, and the $1.00\times$ cancellation measurement—was performed adversarially: every claim was re-implemented from scratch and tested against the hash standard rather than accepted from prior text. The $1.00\times$ result, in particular, was not the expected outcome; the first run (with the answer-key) showed the $5\times$ collapse, and only removing the oracle exposed the cancellation. We report this sequence honestly because the order of discovery is part of the result: the collapse is real, and so is the reason it does not constitute an inversion.

The SHA-256 Round and Its Reverse Step

We fix notation for one block of SHA-256 acting on the eight 32-bit state words (a, b, c, d, e, f, g, h) with round constants K_t and message schedule W_t, $t = 0, \ldots, 63$. Writing Σ_0, Σ_1 for the rotation mixers, Ch and Maj for the Boolean functions, the round is

$$\begin{aligned} T_1 &= h + \Sigma_1(e) + \mathrm{Ch}(e, f, g) + K_t + W_t, \\ T_2 &= \Sigma_0(a) + \mathrm{Maj}(a, b, c), \\ (a', b', c', d', e', f', g', h') &= (T_1 + T_2,\ a,\ b,\ c,\ d + T_1,\ e,\ f,\ g). \end{aligned}$$

The first thing to observe is the *conveyor* structure: six of the eight output words are verbatim copies of input words ($b' = a$, $c' = b$, $d' = c$, $f' = e$, $g' = f$, $h' = g$). Only a' and e' receive freshly computed values. This is the backbone on which everything else rests, and we verified it holds with zero mismatch across all 64 rounds.

Theorem 1 (One-step reverse recovery). *Given the full next-state vector* $(a', b', c', d', e', f', g', h')$ *and a candidate value for the message word* W_t*, there is a unique predecessor state* (a, b, c, d, e, f, g, h) *consistent with the round function, computable in a constant number of word operations.*

Proof. The conveyor copies invert immediately: $a = b'$, $b = c'$, $c = d'$, $e = f'$, $f = g'$, $g = h'$. With a, b, c in hand, $T_2 = \Sigma_0(a) + \mathrm{Maj}(a, b, c)$ is determined, so $T_1 = a' - T_2 \bmod 2^{32}$. Then $d = e' - T_1 \bmod 2^{32}$. Finally, with e, f, g known and W_t supplied as the candidate, $h = T_1 - \Sigma_1(e) - \mathrm{Ch}(e, f, g) - K_t - W_t \bmod 2^{32}$. Every step is a fixed arithmetic or Boolean operation, so the recovery is constant-time. o□ □

Theorem 1 is the heart of why the difficulty does not live in the reverse step. Reversal is a closed-form computation. The catch is the phrase "a candidate value for W_t": the predecessor is unique *given* the word, but the word itself ranges over 2^{32} possibilities, and we need a way to recognize the correct one without already knowing the answer.

The conveyor, verified

Before building the recognition criterion, we pause on the conveyor structure, because it is what makes the cross-round constraint of Section 6 meaningful and because it is independently verifiable to the bit. We ran the forward pass on **b"abc"** and checked, at every round t from 1 to 63, the six copy relations $b_t = a_{t-1}$, $c_t = b_{t-1}$, $d_t = c_{t-1}$, $f_t = e_{t-1}$, $g_t = f_{t-1}$, $h_t = g_{t-1}$. All held with zero mismatch. Only a_t and e_t carry fresh information into each round; the other six words are a delay line. Equivalently, the entire 256-bit state at round t is determined by the two-word "head" (a, e) histories of the preceding three rounds.

This has a direct consequence for reverse exploration. When we recover a predecessor state x_{63} from a candidate W_{63}, six of its eight words are shift-forced by the known next state and do *not* depend on the guess; only d_{63} and h_{63} move with W_{63}. The cross-round constraint of Section 6 therefore bites precisely on these two computed words, while the six conveyor words must *already* be consistent with round 62's output by construction. The conveyor is not a curiosity; it is the rigid scaffold that makes "is this recovered state reachable?" a sharp question.

The forward map has no gradient: why a basin exists at all

It is worth establishing, on the bench, *why* a recognition criterion is needed in the first place—why one cannot simply hill-climb toward the true word. The reason is the avalanche property: a one-bit change in the input produces an essentially independent output, so the map presents no smooth "warmer/colder" signal in the raw bits.

Avalanche measurement over $2,000$ one-bit input perturbations of b"abc". The mean fraction of output bits flipped is 0.5007, indistinguishable from 0.5. The forward map presents no readable gradient: a single-bit nudge to the input randomizes the output, so there is no smooth descent toward the answer in the output bits themselves.

Figure 1 measures the flatness directly: the distribution of output-bit flip fractions is tightly centered on 0.500. This is the precise sense in which naive search has nothing to climb. The geometry-only score of Section 4 is an attempt to manufacture a gradient *where the raw bits offer none*—to read tension in the carry structure rather than warmth in the output. That it succeeds in producing a provable zero at the truth

(Section 5) is genuine; that the zero sits in a wide cold basin (Section 6) is the price of building a gradient on top of a field that is natively flat. The two-read-head experiment then asks whether *depth* can supply the discrimination that a single round's manufactured gradient cannot.

The Geometry-Only Score

The tension probe's key idea is to recognize the true word not by what it *is* but by the carry geometry it *induces*. Each addition in the computation of T_1 is staged, and at each stage we record two things: the carry-out bit (whether the 32-bit addition overflowed) and the Hamming weight of the internal carry mask (how many bit positions generated a carry into the next).

Definition 1 (Geometry bundle). *For round t, the geometry bundle G_t consists of: the four staged carry-out bits of the T_1 additions, the four Hamming weights of the staged carry masks, and the Hamming weight $\mathrm{hw}(h)$ of the incoming word h.*

Definition 2 (Tension score). *Given the next state, a candidate word W_t, and the true bundle G_t exported from the forward pass, recover the predecessor by Theorem 1, recompute the candidate's own staged additions, and define $S_t = 5 \cdot (\text{carry-out mismatches}) + (\text{mask-weight drift}) + |\mathrm{hw}(h_{\text{cand}}) - \mathrm{hw}(h_{\text{true}})|$. Lower is colder. The weight 5 on carry-out mismatches reflects that an overflow-parity disagreement is a coarser structural failure than a one-bit drift in a mask weight.*

The score uses only *geometric invariants* of the addition—overflow bits and carry-density—never a direct comparison of the candidate word to a stored target. In this precise sense it is "geometry only." The score is a tension field: it measures how much the candidate's induced carry structure must deform to match the true structure.

The Zero Score Theorem, Independently Verified

Theorem 2 (Zero Score). *The true message word W_t achieves $S_t = 0$, necessarily, because its recovered predecessor is the true predecessor and therefore reproduces every carry-out bit, every carry-mask weight, and the true $\mathrm{hw}(h)$ exactly.*

Proof. By Theorem 1, supplying the true W_t recovers the true predecessor state. The forward additions of T_1 from the true predecessor are, by definition, the additions whose geometry was exported as G_t. Hence each recomputed carry-out bit matches, each recomputed mask weight matches, and $\mathrm{hw}(h_{\text{cand}}) = \mathrm{hw}(h_{\text{true}})$. All three penalty terms vanish and $S_t = 0$. o☐ ☐

This much is structural and provable. The empirical content is whether the true word is *distinguished* by this zero—whether false words avoid it. We reconstructed the entire probe independently and tested on the canonical input **b"abc"**, whose final-round word is $W_{63} = \texttt{0x12b1edeb}$.

Independent reconstruction of the Zero Score Theorem. The true word scores exactly zero; not one of $200,000$ random false words does.

Candidate	Geometry score S_{63}

Candidate	Geometry score S_{63}
true W_{63} = 0x12b1edeb	**0**
0x00000000	26
0x12345678	31
200,000 random words, count scoring 0	**0**
200,000 random words, count scoring ≤ 1	5

Table 1 confirms the theorem on an independent implementation. The true word is the unique zero in the sampled space, and the gradient is real: false words score high, and the count of near-zero false words is tiny. This is a genuine tension field with a genuine attractor at the truth.

The Cold Basin: Where the Difficulty Reappears

The Zero Score Theorem might suggest the problem is solved: score every word, keep the zero. The difficulty reappears one level down, in the *width* of the cold basin. While the exact-zero set is essentially the singleton truth, the *near*-cold set—words scoring at or below a small threshold—is large.

One-round cold basin size as a function of score threshold, estimated from $200,000$ samples scaled to the full 2^{32} space. The exact-zero basin is a singleton at this scale, but the $S_{63} \leq 4$ basin already holds ~ 4.3 million candidates, and $S_{63} \leq 8$ holds ~ 55 million.

Figure 2 shows the basin growing steeply with threshold. The operational consequence: any practical search uses a threshold (exact-zero testing is itself a search), and at threshold $S_{63} \leq 4$ the basin contains roughly 4.3×10^6 candidates. The true word ranks first, but it sits inside a cold population of millions. *This is the gap, localized.* It is not in the reverse step (closed-form) and not in the recognition of the exact truth (a provable zero). It is in the fact that the one-round criterion is satisfied—to within tolerance—by an exponentially large set of impostors.

The Central Experiment: Two Read-Heads

The first author's standing intuition is that this basin is an artifact of reading only one round, and that *depth*—chaining the criterion across successive rounds—collapses it. The round function's conveyor structure gives this teeth: a candidate W_{63} recovers a full predecessor state x_{63}, and x_{63} was itself produced by round 62. So x_{63} is not free; it must be *reachable*, and reachability is a cross-round constraint.

We make this precise with two read-heads on the same structure.

Definition 3 (Two read-heads). *Read-head A (forward verifier) scores a candidate W_{63} at round 63. Read-head B (depth) takes the predecessor x_{63} recovered from a round-63-cold candidate and scores it at round 62. A candidate is two-round cold if it passes both heads.*

The decisive design choice is *what read-head B is allowed to know*. There are two versions, and the difference between them is the entire result.

Version 1: with the answer-key

Here read-head B scores x_{63} at round 62 using the *true* word W_{62}. This is an answer-key verifier: it assumes the predecessor word is known and asks only whether the recovered x_{63} is consistent with it.

Version 1 (answer-key). One added round collapses the basin ≈ $5\times$.

	round-63 cold (≤ 4)	two-round cold (≤ 4)
sample of 500,000 words	495	107

Table 2 shows a clean ≈ $5\times$ collapse per added round. Extrapolating geometrically, $495 \cdot (107/495)^k = 1$ gives $k \approx 4.1$: about four more rounds of this constraint would isolate the true word. *If* the predecessor words were known, the depth axis would be ferocious, and the basin would close in a handful of rounds. This is the first author's intuition, vindicated—under an assumption.

Version 2: without the answer-key

The assumption is exactly the thing an inversion does not get. In a real reverse exploration you do not possess W_{62}; you must *find* it, by the same cold criterion, searching its own 2^{32} space. Read-head B is now honest: for each round-63-cold candidate, recover x_{63}, then *search* the W_{62} space for a word that scores cold at round 62.

Version 2 (honest). Every round-63 cold seed extends to a round-62 cold seed. The basin does not collapse: the per-round collapse factor is exactly $1.00\times$.

Measurement	Value
round-63 cold seeds tested	300
seeds admitting a cold W_{62} (in 20,000 samples each)	**300**
fraction extending to round 2	**1.000**
1-round cold density d_1	1.085×10^{-3}
fraction extending	1.000
2-round joint density d_2	1.085×10^{-3}
density collapse per added round	**$1.00\times$**

Table 3 is the central result. *Every single* round-63 cold seed (300 of 300) admits a cold predecessor word. The basin does not shrink. The density collapse factor is $1.00\times$. The $5\times$ collapse of Version 1 was *entirely* the contribution of the answer-key: it was the true W_{62} doing the work, not the geometry. Remove the oracle, and the added round contributes exactly as much new freedom (its own cold W_{62} choices) as it removes by constraint.

The two read-heads. With the answer-key (blue), one added round collapses the basin $5\times$. Without it (red), the basin holds flat at $1.00\times$. The collapse is the oracle, not the geometry.

Interpretation: The Cancellation as Local Conservation

The two-read-head structure of a single reverse round. Read-head A tightens (the basin collapses with depth); read-head B widens (each round needs its own word). At the seam where they meet, collapse $\times$ cost $= \mathbf{1.00}\times$.

The two read-heads are not two different experiments giving conflicting answers. They are two measurement points on one structure, and the structure has a genuine two-sidedness. Reading from the answer-key side, depth is a powerful tightening force. Reading from the honest side, depth is self-funding: it pays for its own constraint with its own unknown.

We propose the following reading, and we are careful to label it as interpretation rather than theorem. The quantity that the depth axis collapses and the quantity that the search axis inflates are, in this regime, the same quantity. The constraint that would collapse the frontier (knowledge of the predecessor word) is exactly as expensive to obtain as the collapse is valuable. This is a measured, local instance of a conservation phenomenon: the difficulty is not removed by adding depth; it is *relocated* from the width of one basin into the search for the next round's word, in equal measure. The $1.00\times$ is the signature of that equality.

Observation 1. *The $1.00\times$ cancellation is consistent with, but does not prove, a global conservation law. It establishes the equality for the two-round, geometry-only case on one input. Whether the equality is exact across all 62 available rounds, across inputs, and under richer cross-round constraints than the single bundle used here, is open.*

What Would Break the Cancellation

Honesty about a null-shaped result requires naming what would overturn it. The $1.00\times$ figure would be broken—and the first author's collapse intuition would be vindicated without the oracle—by any of the following, demonstrated on the bench:

1. **A super-additive cross-round constraint.** Our read-head B uses the same single geometry bundle at round 62 that it uses at round 63. A constraint that couples rounds 62 and 63 *jointly*—using information from the round-63 recovery to restrict the round-62 search below its independent cold density— could yield an extension fraction strictly below 1.0, hence a collapse factor above 1.0. We did not find such coupling in the single-bundle score; whether a richer bundle exhibits it is the natural next experiment.
2. **A schedule constraint.** The words W_t are not independent for $t \geq 16$; they are determined by the message schedule recurrence. For the last 48 rounds, a cold W_{62} that is *inconsistent with the schedule* given the recovered neighbors is spurious. Folding the schedule recurrence into read-head B is a concrete, runnable way to shrink the extension fraction, and is the most promising route to a genuine sub-1.0 collapse.

3. **A basin whose exact-zero set chains.** Our cancellation uses the ≤ 4 cold basin. The exact-zero set is far smaller (a near-singleton per round). If exact-zero predecessors chain with extension fraction below 1.0, the picture changes—though exact-zero search is itself expensive, which is the conservation reasserting itself one level up.

Each of these is a falsifiable, bench-ready experiment. We report the $1.00\times$ result as the honest current state, not as a closed verdict.

The Backward Wedge: What the Digest Alone Forces

The two-read-head experiment concerns recovery of message words given a full intermediate state. A separate and complementary question is how much of the internal trajectory is forced by the *digest alone*—the eight output words a verifier actually possesses. This bounds, from a different direction, how much "free" structure reverse exploration inherits before any search begins.

Recall that the digest is the feed-forward sum of the initial state and the final round state. Running the recovery of Theorem 1 in the feed-forward direction, we find that a substantial block of the final rounds is *forced*: given the digest, the conveyor copies and the feed-forward relations determine 28 of the schedule-region words—896 bits—without any guessing. We verified this forcing directly: all 28 forced words, recomputed from the digest, matched the true forward computation exactly (28/28).

The shape of this forcing is geometric, and it is worth describing because it clarifies what reverse exploration is up against. The forced region is a *right triangle* in the round–word plane: near the output, many words are pinned by the digest; moving backward into the computation, the number of pinned words shrinks linearly until, past a certain depth, nothing is forced and the full search space opens. The hypotenuse of this triangle is the boundary between the inherited (forced, free) structure and the sealed interior where the $1.00\times$ cancellation governs.

This backward wedge is not in tension with the cancellation result; it complements it. The wedge says a fixed, $O(1)$-sized slab of the trajectory is free—the verifier gets the last several rounds' worth of structure for nothing. The cancellation says that *beyond* the wedge, where genuine search begins, adding depth does not reduce the search, because each round's freed constraint is matched by its own freed unknown. Together they give a complete qualitative picture of single-block reverse exploration: a free triangular wedge at the output, then a sealed interior where depth is self-funding. Neither the wedge nor the cancellation, alone or together, constitutes an inversion; the wedge is finite and the interior is balanced.

The two-read-head cancellation has a complementary, and arguably more fundamental, explanation in terms of *how many unknown channels* a reverse step must resolve at once. We make this precise with a minimal model that strips away everything but the addition structure, following the hot/cold nibble construction .

Consider recovering an unknown word X from a known sum $T = S + X \pmod{2^{32}}$, worked one nibble (four bits) at a time with carry propagation. At each nibble, given the target nibble, the known nibble, and the incoming carry, the unknown nibble is *forced*: there is exactly one value of X's nibble that produces the correct target nibble and a determinate carry-out. We verified this: every nibble is unique, the per-nibble "heat" (the base-2 logarithm of the number of admissible continuations) is 0, and the word crystallizes upward with no search. This is the *cold* regime—one unknown channel, zero heat, unique recovery.

Now consider $T = S + X + Y \pmod{2^{32}}$ with *two* unknown channels. At each nibble, the number of admissible (X-nibble, Y-nibble,carry-out) triples consistent with the target jumps to 16. The per-nibble heat is $\log_2 16 = 4$ bits, and over eight nibbles the frontier carries 32 bits of branching. This is the *hot* regime—two unknown channels, four bits of heat per nibble, an open search frontier.

Channel counting. One unknown channel ($T = S + X$, green) forces a unique nibble at every position: heat 0, crystallization, no search. Two unknown channels ($T = S + X + Y$, red) admit 16 continuations per nibble: heat 4 bits, a branching frontier. The transition from cold to hot is exactly the transition from one unknown to two.

Figure 5 shows the regimes side by side. This is the microscopic engine of the $1.00\times$ cancellation. Read-head A, recovering W_{63} against a fully known next state, is closer to the one-channel (cold) case—the predecessor is forced *given* the word, and the word is recognized by its zero. But the moment read-head B must also find W_{62}, the reverse step is resolving *two* unknown words against the cross-round constraint, and the second channel re-injects exactly the branching that depth was supposed to remove. The hot/cold model predicts the cancellation qualitatively: *adding a round adds a channel*, and a fresh unknown channel contributes heat at the same rate the cross-round constraint removes it. The $1.00\times$ figure is the quantitative shadow of one-channel-in, one-channel-out.

We emphasize what this model is and is not. It is an honest thermometer: it *measures* which regime a reverse step occupies and how much heat it carries. It is not a refrigerator: it does not move a two-channel problem into the one-channel regime. The genuine contribution of the hot/cold construction is that it makes the readable/sealed distinction *quantitative and local*— a per-nibble heat reading—rather than a global verdict. A structure is cold where it presents one unknown channel and hot where it presents two or more, and the SHA-256 reverse exploration is hot precisely because honest depth forces a second channel at every round.

We are deliberately cautious here. Nothing in this paper proves or disproves any separation of complexity classes, and we make no such claim. What the experiment provides is a concrete, measured picture of *where* the difficulty of reverse exploration lives in one well-studied function:

- not in the reverse step (Theorem 1: closed-form, constant-time);
- not in recognizing the exact truth (Theorem 2: a provable zero, empirically unique in samples);
- but in the *width of the cold basin* and, decisively, in the fact that the cross-round constraint which would close that basin is itself a search of equal cost (Table 3: $1.00\times$).

The two-read-head framing makes the structure legible: forward verification is cheap, backward depth is the lever, and the lever's fulcrum is the cost of the predecessor word. The honest measurement is that, with the constraints we tested, the lever is exactly balanced. This is a far sharper statement than "SHA-256 is hard to invert." It says *which* balance holds, *where*, and *by how much*—and it hands the next researcher three specific ways to try to tip it.

The Message Schedule: The Most Promising Route to Sub-Unity

Of the three falsifiable routes named in Section 9, the message schedule deserves expanded treatment, because it is the one place where the *independence assumption* underlying the $1.00 \times$ result is known to be false, and where a genuine sub-unity collapse is most plausible.

The two-read-head experiment of Section 6 treats W_{63} and W_{62} as free 32-bit unknowns, each searched over its own 2^{32} space. For the first sixteen rounds this is exactly right: $W_0, \ldots, W_{15}$ are the raw message block and are genuinely independent. But for $t \geq 16$ the schedule is generated by the recurrence

$$W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}},$$

where σ_0, σ_1 are the schedule's small rotation-and-shift mixers (distinct from the round mixers Σ_0, Σ_1). Both W_{62} and W_{63} lie deep in the recurrence-generated region. They are not independent draws; each is a fixed function of six earlier words, and consecutive schedule words share inputs.

This matters for read-head B in a specific way. In the honest experiment, a candidate W_{62} counts as "cold" if it scores low on the round-62 geometry bundle. But a schedule-aware read-head would impose an additional, far more restrictive test: W_{62} must be *consistent with the recurrence* given its neighbors. Most of the cold-scoring W_{62} values that inflated our extension fraction to 1.000 are almost certainly *schedule-spurious*— they produce acceptable carry geometry but could never arise from any actual message, because no message block generates them at position 62.

We did not implement the schedule constraint in the present experiment, and we are explicit that this is the principal limitation of the $1.00 \times$ result: it measures the cancellation in the *geometry-only*, *schedule-free* model, where the words are treated as independent. Folding the recurrence into read-head B is a concrete and runnable next step, and it is the experiment most likely to yield an extension fraction strictly below 1.0, hence a real cross-round collapse without any answer-key.

There is, however, a conservation-flavored caution even here, and intellectual honesty requires stating it. The schedule recurrence constrains W_{62} in terms of $W_{60}, W_{55}, W_{47}, W_{46}$ (the words at offsets 2,7,15,16 back). To *use* that constraint, read-head B must know or jointly solve for those earlier words—which are themselves unknowns deeper in the same reverse exploration. So the schedule may not simply hand over a free collapse; it may relocate the cost again, from "search W_{62} freely" into "jointly satisfy a coupled system of schedule equations across many rounds." Whether that coupled system is easier than the independent search—whether the schedule is a net constraint or merely a constraint that drags its own unknowns along—is, precisely, the open question. Our prediction, which we flag as a prediction and not a result, is that the schedule produces a real but *bounded* collapse: sub-unity for a few rounds, reverting toward unity as the coupled unknowns accumulate. Only the bench will say.

The Dual-Wave Reading: Where You Drop the Read-Head

We close the technical development with the framing that motivated the experimental design, stated plainly as an interpretive lens rather than a mathematical claim.

The two read-heads are not a metaphor laid over the data; they are two literal measurement procedures applied to one structure, and the structure genuinely answers differently depending on which procedure you run. Drop the read-head on the forward side—ask "does this word score cold at this round?"—and you see a tightening: depth, via the answer-key, collapses the basin $5\times$ per round (Table 2). Drop the read-head on the honest backward side—ask "can each round find its own cold word?"—and you see no tightening at all: $1.00\times$ (Table 3). Both readings are correct. They are the same fold viewed from its two faces.

This is what we mean by a dual-wave structure: the collapse and the cost are not sequential events (first the basin tightens, then later you pay for it) but *simultaneous faces* of one operation. Every added round of depth is, at once, a constraint that tightens the basin and an unknown that widens the search, and in the geometry-only model these two faces are exactly equal. The $1.00\times$ is the measured signature of that equality—the point on the seam where the tightening wave and the widening wave have the same amplitude and cancel.

We are careful about what this does and does not settle. It does *not* settle any question about complexity classes; we have proven nothing about P and NP, and we claim nothing about them. What it settles, on the bench, for this function and this model, is a local fact with a clean number: the difficulty of reverse exploration is conserved across depth. It is not destroyed by chaining rounds—the intuition that depth alone collapses the basin is false once the answer-key is removed—and it is not amplified either; it is *relocated*, in exactly equal measure, from the width of one round's basin into the search for the next round's word. A quantity that moves but neither grows nor shrinks under a transformation is, in the ordinary sense of the word, conserved. We have measured that conservation locally and named the one assumption (word independence) whose removal could break it.

There is a discipline in this result that we want to make explicit, because it is the discipline that produced it. The $5\times$ collapse was the seductive number—it pointed toward inversion, and had we stopped there we would have reported a breakthrough. The $1.00\times$ is the true number, and it required removing a crutch we had not initially noticed we were leaning on. The lesson is the one the first author's method insists upon: the clean, collapsing, inversion-shaped result was the warning, not the verification; the verification was in the resistance—the place where, once the answer-key was removed, the structure refused to collapse. The honest headline of this paper is the number that resisted.

Conclusion

We reconstructed a geometry-only reverse tension probe for single-block SHA-256 on an independent bench, re-derived and confirmed its Zero Score Theorem (the true word, and empirically only the true word, scores exactly zero), and then resolved the open question of cross-round basin collapse with a direct measurement. The answer is two-faced and clean. With an answer-key, depth collapses the cold basin $\approx 5\times$ per round; without one, the collapse is exactly $1.00\times$—each added round of depth tightens the basin precisely as much as the search for that round's own word widens it. We read this as a local conservation: the difficulty of reverse exploration is not destroyed by depth but relocated, in equal measure, from basin width into per-round search.

The result is novel in three respects: the independent confirmation of the Zero Score Theorem; the explicit two-read-head experimental design that separates the oracle's contribution from the geometry's; and the $1.00\times$ cancellation measurement, which we believe has not previously been isolated. It is also, emphatically, *not* an inversion of SHA-256, and the honest headline is the $1.00\times$, not the $5\times$. We have named three falsifiable experiments— joint cross-round coupling, the schedule recurrence, and exact-zero chaining— any of which could overturn the cancellation and which constitute the natural continuation of this work.

Methods and Honest Limitations

We collect here the methodological choices and limitations, both so the result can be reproduced and so its boundaries are unambiguous.

Scoring choices.

The geometry score weights carry-out mismatches by 5 and mask-weight drift and $\mathrm{hw}(h)$ drift by 1. These weights were not tuned to produce the result; they encode the qualitative judgment that an overflow-parity disagreement is a coarser structural failure than a single-bit drift in carry density. We checked that the Zero Score Theorem (Theorem 2) is insensitive to the weights—the true word scores zero under any nonnegative weighting, because all penalty terms vanish simultaneously at the truth. The cold-basin *width*, and hence the absolute survivor counts, do depend on the threshold and weights; the $1.00\times$ cancellation, being a *ratio* of densities at a fixed threshold, is robust to these choices in our checks.

Sampling.

All densities were estimated by seeded pseudorandom sampling of the 2^{32} word space, with sample sizes from 2×10^5 to 5×10^5 for one-round densities and 300 seeds (each with 2×10^4 inner samples) for the two-round extension fraction. The extension fraction of 1.000 $(300/300)$ is a strong signal but a finite sample; we do not claim it is provably exactly 1, only that no seed in the sample failed to extend, which already suffices to refute a rapid collapse.

Single input.

All experiments used the canonical input **b"abc"**. The Zero Score Theorem is input-independent (it is structural). The cancellation was measured on one input; we expect it to hold across inputs by the same channel-counting argument, but we did not sweep inputs, and that sweep is part of the natural continuation.

The central limitation.

The $1.00\times$ result is measured in the *geometry-only*, *schedule-free* model, treating W_{62} and W_{63} as independent 32-bit unknowns. As Section 10 details, the message schedule couples these words, and a schedule-aware experiment could yield a sub-unity collapse. We therefore state the result with its scope attached: *under independent per-round word search with the single-bundle geometry score, the cross-round cold-basin collapse factor is* $1.00\times$*.* Removing the independence assumption is the foremost open experiment, and we have predicted (Section 10) a bounded sub-unity collapse rather than a full one—a prediction the bench can confirm or refute.

What we did not do.

We did not attempt a full multi-round chained recovery to a unique word, because the honest two-round measurement already shows the per-round driver is balanced; chaining a balanced step does not concentrate. We did not implement the schedule recurrence, exact-zero chaining, or joint cross-round bundles—these are the three named falsifiable continuations. We make no claim about SHA-256's cryptographic security, which is untouched by any result here.

Every number in this paper was produced by direct computation against the SHA-256 standard. The forward pass, the conveyor verification, the reverse recovery (Theorem 1), the Zero Score Theorem test (Table 1), and both versions of the two-read-head experiment (Tables 2, 3) were run on the canonical input **`b"abc"`** with seeded pseudorandom sampling. The true final word is $W_{63} = \mathtt{0x12b1edeb}$; the true penultimate word is $W_{62} = \mathtt{0xeeaba2cc}$; the digest is **`ba7816bf8f01cfea...`**. Independent re-implementation from this description should reproduce the $1.00\times$ cancellation.

9 D. A. Kulik, *A Geometry-Only Tension Probe for Reverse Exploration of Single-Block SHA-256*, Ver. Mark 9, QuHarmonics Research Group, 2026. National Institute of Standards and Technology, *Secure Hash Standard (SHS)*, FIPS PUB 180-4, 2015. D. A. Kulik, *Understanding the Nexus Lens: Hot/Cold Nibble Runtime*, QuHarmonics Research Group notebook, 2026.